A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Captive breeding, habitat creation & management - Set 2

Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **In-situ:** species remain in natural conditions with the whole community and inter-species relationships – protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • **Ex-situ:** captive breeding when in-situ is impossible or insufficient – selection criteria – difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation) – techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning) – seed banks – successes (Hawaiian goose, Iberian lynx, red kite, large blue) • **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to – both needed, in-situ fundamental **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	9
	• **plagioclimax** → a community held back from its climax by human activity **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented.	
	Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring.	
	The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is.	
	Weaker answer - Level 2 (5 marks)	
	In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together.	
	What makes the difference	
	• Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples.	
	• Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only.	
	• 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most.	
01.2	1 mark for the point below. **1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** A larger area supports larger populations, which are less likely to be lost to chance events or inbreeding, and it can hold species with large territories such as top predators **1** A larger area holds a wider range of habitats and microhabitats, so more species can find the conditions they need **1** A compact, rounded shape has the smallest perimeter for its area, so less of the reserve is affected by edge effects such as wind, drier conditions, spray drift and disturbance from outside **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** It allows individuals to move between the two areas, so the populations interbreed as one larger gene pool and inbreeding is reduced **1** An area that loses a species can be recolonised from the other, and species can shift their range as the climate changes rather than being trapped in an isolated fragment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
02.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water levels:** sluices and bunds hold water back to create ponds, scrapes and seasonally flooded marsh, adding wetland habitats and a range of depths for different species **1 Soil nutrient levels:** cutting and removing hay rather than leaving it, and never fertilising, lowers the nutrient level so that vigorous grasses cannot out-compete the wildflowers of species-rich grassland **1 Topography:** banks, scrapes and south-facing slopes create warm and cool, wet and dry microhabitats within a small area, each suiting different species **1 Light levels:** thinning, coppicing and cutting rides and glades lets light reach the woodland floor, so ground flora and the insects that depend on it can grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	4

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** A plagioclimax is a community held at an earlier stage of succession than its natural climax by continued human activity, so it never develops into woodland **1** Grazing animals eat the tree and shrub seedlings that would otherwise shade out the grassland, so succession is deflected and the open habitat survives **1** The species, stocking density and timing of grazing are chosen to suit the habitat – cattle graze less selectively and their trampling creates bare patches for seedlings, while sheep graze close and produce a short, even sward **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
02.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invasive non-native species such as rhododendron or Himalayan balsam out-compete and shade out native species, and have no natural controls here **1** Introduced predators or competitors, such as grey squirrels or mink, reduce native populations directly **1** Without their natural predators, herbivore numbers such as deer can exceed the carrying capacity, preventing tree regeneration and damaging the ground flora **1** Culling may also be needed to keep a population within the reserve's carrying capacity so that it does not damage the habitat the whole community depends on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-49; AQA spec 3.1.3.3-3.1.3.4.	

Do not write outside the box